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FINAL PROJECT REPORT

ARTIFICIAL INTELLIGENCE AND ADVANCED LARGE LANGUAGE MODELS

PROJECT / RESEARCH TITLE

Neural Fields for Continuous Cardiac Cine-MRI

One heartbeat stored as one continuous function f(x, y, z, t): experiments, a 4D heart model and its clinical check

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Abstract

A cardiac cine-MRI scan stores one heartbeat as a grid: in our main patient 224 × 256 pixels of 1.64 mm, 10 slices 10 mm apart and 30 frames, 17.2 million values with nothing stored between slices or between frames. We represent the scan instead as one continuous function, a small sine-activated network (SIREN, 198,913 weights) that maps a position in millimetres and a moment of the heartbeat to image intensity, and we fit it per patient on the public ACDC dataset using a laptop CPU. We tested it honestly: slices or frames were hidden during training and then predicted, against linear interpolation and against a ReLU network of the same size. Three findings stand out. First, SIREN is 6.6 dB better than the ReLU network on hidden frames (28.9 vs 22.3 dB PSNR in the heart). Second, how the coordinates enter the network matters as much as the network itself: scaling the time coordinate (t × 0.1) and the slice coordinate (z × 0.25) inside the model removes the oscillation between known samples and adds 6.6 dB in time. Third, the field is level with linear interpolation (28.9 vs 29.2 dB with every fourth frame kept), better during the fastest contraction, and a fixed 50/50 blend of both beats interpolation (29.7 dB, 18 of 21 frames). The same weights hold the heart region with 16× fewer numbers, or the whole scan with 86× fewer at 27.5 dB. From the final field we extract a beating 3D heart on a 1.64 mm isotropic grid; its ejection fraction (LV 49 %, RV 51 %) is within five points of the expert labels (44 %, 53 %).

1. Objectives

The proposal set four objectives, all addressed in this report: (O1) a reproducible pipeline from an ACDC NIfTI scan to coordinates in physical units; (O2) a SIREN fit of a static 3D frame, tested on hidden slices against interpolation and a ReLU control; (O3) a 4D fit of the whole cycle with three time encodings, tested on hidden frames; (O4) a demonstration of the continuous cardiac cycle. During the work we added two items: a measurement of the compression–quality trade-off promised at the midterm (about 86×), and a clinical check of the 4D model through chamber volumes and ejection fraction.

2. Data

We use the ACDC dataset (MICCAI 2017) [3]: cine-MRI of 150 patients with expert labels of the right ventricle, the myocardium and the left ventricle at end-diastole (ED) and end-systole (ES). The main patient is 051 (previous myocardial infarction): 224 × 256 pixels of 1.64 × 1.64 mm, 10 short-axis slices 10 mm apart, 30 frames, ED = frame 1, ES = frame 11. Patient 015 (dilated cardiomyopathy; 256 × 216 pixels of 1.37 mm, 9 slices 10 mm apart, 21 frames) is used to repeat the hidden-slice test on a different heart. Both patient folders are submitted with this report together with the citation required by the dataset.

3. Method

3.1 Representation

Every voxel becomes a sample $(x, y, z, t) \rightarrow I$. Spatial coordinates are taken at voxel centres in millimetres, centred and divided by half of the largest physical extent, so the true 1.64 : 10 mm ratio is kept; $t = 2k/(T - 1) - 1$ for frame k of T; intensities are min–max scaled to $[-1, 1]$. The field is a SIREN [1]: four layers of 256 units computing $\sin(\omega_0(Wx + b))$ with $\omega_0 = 30$ and the initialisation of the original paper, and a linear output. The control network has the same layer sizes with ReLU activations (also 198,913 weights).

3.2 Our contribution: scaling the coordinates inside the model

With $\omega_0 = 30$, a small step in a coordinate already turns the first-layer sines by radians. Trained on frames (or slices) that are far apart in that coordinate, the network memorises them and oscillates in between. We therefore multiply the time coordinate by a factor s and, for the slice test, the z coordinate by a factor s_z

inside the model, which slows the field down along that axis only. For time we compared raw t, scaled t and a periodic encoding ($s \cdot \sin \varphi$, $s \cdot \cos \varphi$) of the cardiac phase $\varphi = 2\pi k/T$, and swept s; for z we swept s_z.

3.3 Training

Mean squared error, Adam, learning rate 10^{-4} with cosine decay to 10^{-5}, random batches of 16,384 points. To keep CPU training practical the field is fitted inside a square in-plane crop around the expert-labelled heart (104 × 104 pixels, 12-pixel margin); normalisation always refers to the whole volume. The 4D runs use 20,000 steps (about 25 minutes on an Intel Core Ultra 5 laptop CPU without a GPU); the final field is trained the same way on all slices and all frames.

3.4 Evaluation protocol

Only data hidden from the network is scored. Hidden slices: every second slice of the ED frame (slices 1, 3, 5, 7), so the known slices are 20 mm apart. Hidden frames: every second frame, or, in the harder setting, only every fourth frame kept (frames 0, 4, …, 28, 29; 21 hidden). Baselines: linear interpolation between the nearest known slices or frames (it uses exactly the same known data) and the ReLU control. Metrics: PSNR with a data range of 2 inside the heart (expert labels of ED and ES, dilated by 5 pixels), PSNR over the crop, and SSIM. A blend 0.5 × field + 0.5 × interpolation was also scored; its weight was fixed before testing and never tuned on hidden data.

3.5 From the field to a 3D/4D heart

The final field is queried on an isotropic grid (61 slices of 1.64 mm instead of 10 of 10 mm) at any number of time points. Blood is bright and muscle dark in cine-MRI, so the blood threshold is the midpoint of the median LV-blood and median-myocardium intensities at ED. The expert labels only select the region of each ventricle (interpolated between slices by signed distance); inside it the largest bright 3D component is the blood pool. The LV wall is the surrounding voxels in the muscle intensity band. Surfaces are extracted with marching cubes [8] and smoothed with Taubin smoothing [9]; volumes give the ejection fraction EF = (EDV − ESV)/EDV.

4. Implementation

The code (Python, PyTorch, NumPy, nibabel, scikit-image) is organised as a package with data loading, models, training, evaluation and visualisation modules, plus scripts: run_experiments.py (all experiments), evaluate_blend.py, render_4d.py (3D/4D heart) and pipeline.py, which runs the whole study for any patient folder or any 4D NIfTI file with one command. A setup script creates a virtual environment, installs the dependencies and runs the unit tests (models, sampling, metrics, baselines, a short training run). Each run stores its table, figures, the training curve and a JSON record of all settings and software versions. Repository: github.com/ultralightbeat/cardiac-inr.

5. Results

5.1 Time encoding and scale

Table 1. Hidden frames of patient 051, PSNR in the heart (dB) and SSIM.

Setting	Heart PSNR	Crop PSNR	SSIM
Every 2nd frame hidden · raw t	22.74	23.52	0.543
Every 2nd frame hidden · periodic (sin, cos) × 0.25	28.85	30.61	0.858
Every 2nd frame hidden · scaled t × 0.25	29.30	31.74	0.888
Every 4th kept · t × 0.25	28.20	32.05	0.903
Every 4th kept · t × 0.1	**28.87**	**32.83**	**0.920**

Every 4th kept · t × 0.05	28.80	32.83	0.920
Every 4th kept · t × 0.25, 512 neurons (791,041 weights)	28.92	32.93	0.919

How time enters is decisive: scaling t adds 6.6 dB over raw t, and the periodic encoding is close but slightly worse. With fewer frames the best scale is 0.1. A four times larger network reaches the same quality as the 256-unit network with the better scale, so capacity is not the limiting factor (rows 4–7 used 20,000 steps, rows 1–3 4,000).

5.2 SIREN against interpolation and ReLU

Table 2. Every 4th frame kept (21 hidden frames). Compression = values represented / weights.

Method	Heart PSNR	Crop PSNR	SSIM	Compression
Linear interpolation	29.23	33.74	0.941	none (all kept frames)
SIREN, t × 0.1 (heart crop)	28.87	32.83	0.920	16× (8× in bytes)
ReLU, same size	22.28	24.42	0.562	16×
Blend 0.5 SIREN + 0.5 interp.	**29.65**	**33.92**	**0.940**	**none**
SIREN, whole scan, no crop	27.49	34.04*	0.937	86× (43× in bytes)

* over the whole 224 × 256 image, where the static background inflates PSNR; compare the heart column. With every 2nd frame hidden: interpolation 33.14 dB, SIREN 31.35 dB, blend 32.92 dB.

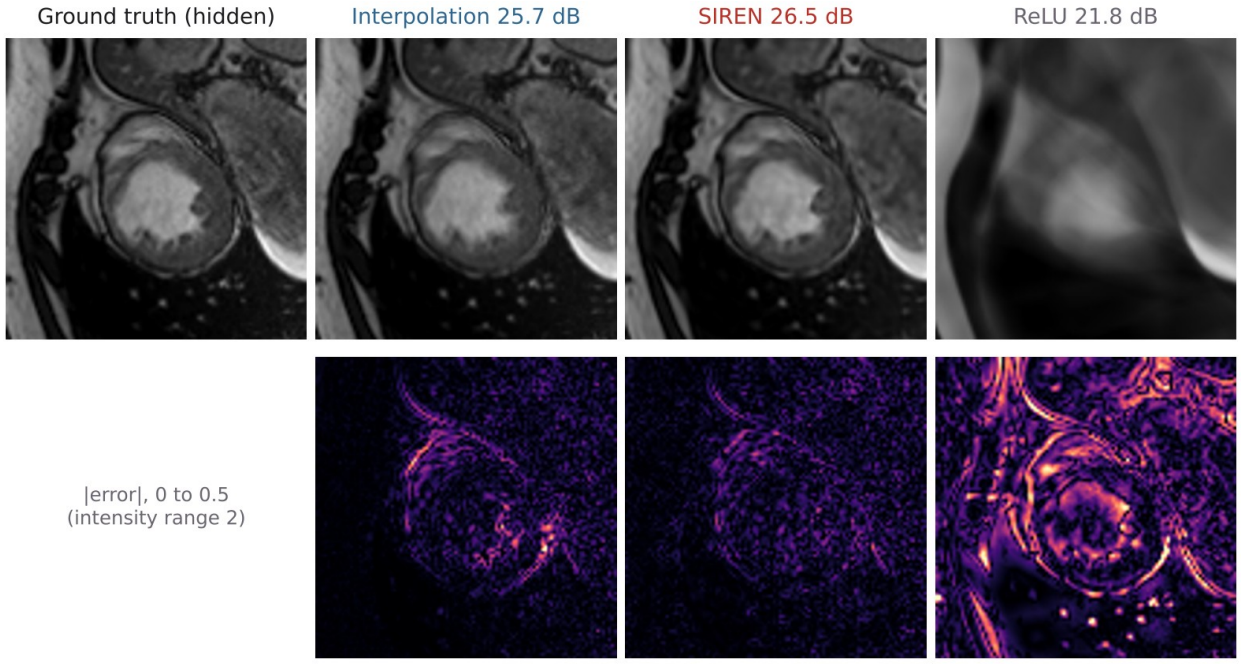


Figure 1. Hidden frame 6 (fast contraction), mid-ventricle slice, every 4th frame kept. Top: prediction and PSNR in the heart for this frame; bottom: absolute error. The ReLU network cannot represent fine detail.

SIREN beats the ReLU network by 6.6 dB on hidden frames; the ReLU network reaches only 24.5 dB even on its own training frames (SIREN 36.5 dB), the spectral bias described in [1, 2]. Against interpolation the field is 0.36 dB behind on average. Per frame (Figure 2) the field wins frames 5–7, the fastest part of the contraction, where a straight line between two frames misses the motion. The two methods fail in different places, and the fixed blend beats interpolation on 18 of 21 frames (+0.42 dB). With dense frames (every second hidden) there is no gain: interpolation is already near the ceiling.

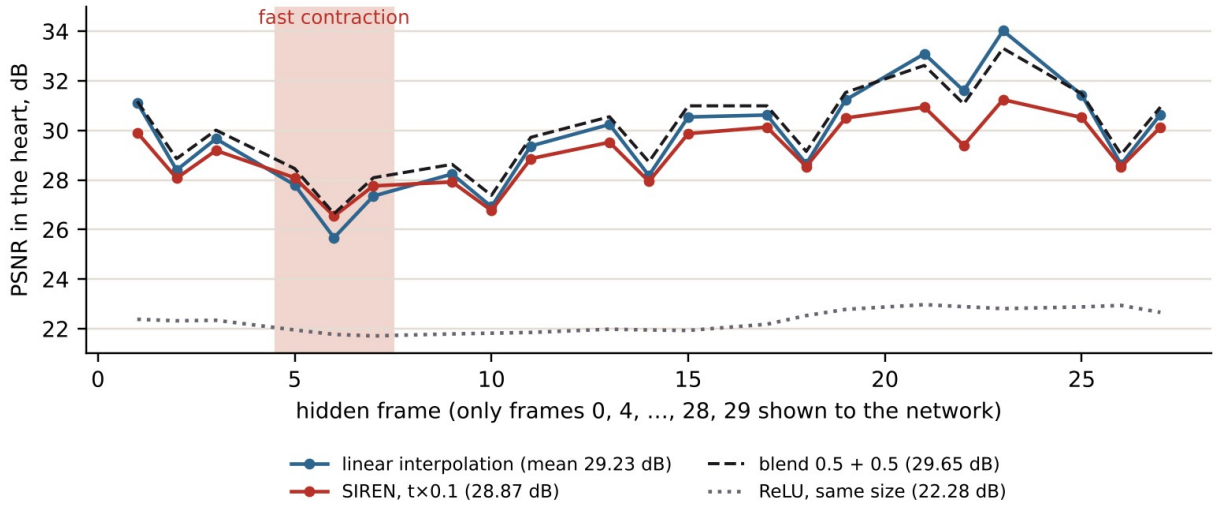


Figure 2. PSNR in the heart for each hidden frame (every 4th frame kept).

5.3 Compression

The field has 198,913 weights. The heart crop holds 104 × 104 × 10 × 30 = 3.24 million values (16× more; 8× in bytes, since voxels are 16-bit and weights 32-bit), the whole scan 17.2 million (86×; 43× in bytes). Fitting the whole scan with the same weights reaches the 86× midterm target at a cost of 1.4 dB in the heart compared with the crop (27.5 vs 28.9 dB). Interpolation is 29.2 dB but compresses nothing.

5.4 Hidden slices

Table 3. Every second slice hidden (20 mm between known slices), ED frame.

Method	Heart PSNR	Crop PSNR	SSIM
Patient 051 · linear interpolation	20.01	19.55	0.354
Patient 051 · SIREN, raw z	19.95	19.72	0.340
Patient 051 · SIREN, z × 0.5	19.87	19.85	0.378
Patient 051 · SIREN, z × 0.25	**20.09**	**19.79**	**0.362**
Patient 051 · ReLU, same size	18.83	19.67	0.373
Patient 015 · linear interpolation	20.68	20.06	0.403
Patient 015 · SIREN, z × 0.5	20.42	20.84	0.410
Patient 015 · SIREN, z × 0.25	20.13	20.45	0.411

No method recovers a slice 10 mm away from both neighbours (all near 20 dB): over 20 mm the heart changes shape. With raw z the field memorised the known slices (41 dB) and filled the gap with high-frequency noise (Figure 3); z × 0.25 removes the noise and makes the field level with interpolation on patient 051. On patient 015 the field is better on the crop and slightly worse in the heart.

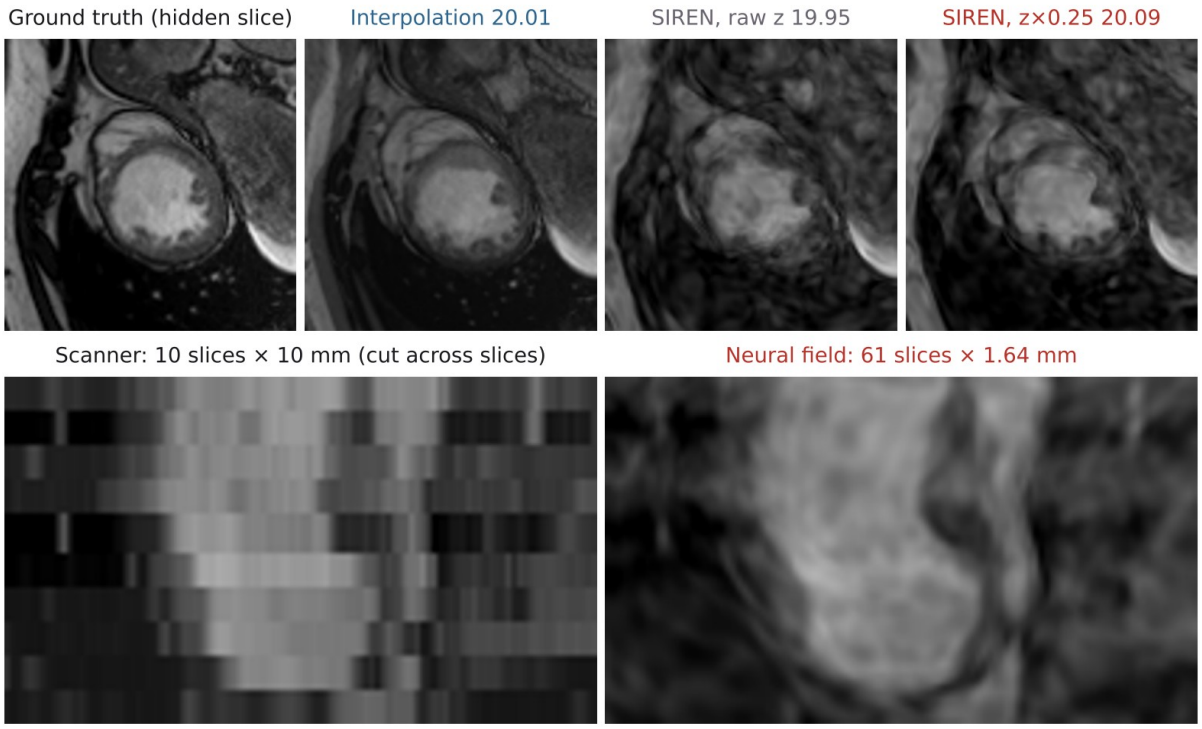


Figure 3. Top: hidden slice 5 and heart PSNR over all hidden slices. Bottom: a cut across the slices of the ED frame, scanner (10 slices) and the final field queried at 61 positions.

5.5 The 4D heart and its clinical check

Table 4. Volumes (ml) and ejection fraction, patient 051.

Source	LV EDV	LV ESV	LV EF	RV EDV	RV ESV	RV EF
Expert labels (ED, ES)	212	120	44 %	142	67	53 %
Our method on scanner slices	176	93	47 %	129	61	53 %
Our method on the neural field	**179**	**92**	**49 %**	**125**	**62**	**51 %**

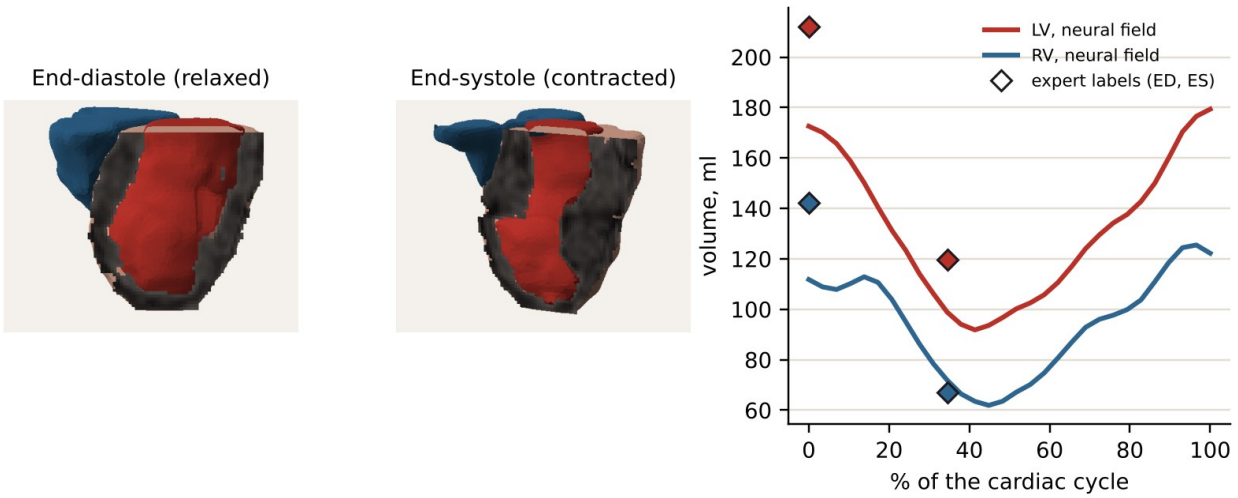


Figure 4. 3D heart from the final field (LV wall cut open: the cut face shows the field's own intensity; red: LV blood; blue: RV blood) and chamber volumes over the heartbeat. An interactive version (heart_4d.html) is part of the submission.

The ejection fractions from the field are within five points of the expert values. Absolute volumes are lower because the intensity threshold leaves the papillary muscles and trabeculae out of the blood pool, while the expert labels include them in the cavity; the ratio is affected much less.

6. Discussion and limitations

- Cine-MRI with 30 frames is close to linear between neighbouring frames, so linear interpolation is a strong baseline; the field gains where the motion is fastest and when frames are sparse.
- The field is a compact, continuous and differentiable representation: one 0.8 MB file of weights per heartbeat that can be queried anywhere. The blend gives the best reconstruction but needs the stored frames.
- Limitations: one main patient and one check patient; a 20 mm slice gap is beyond any method; on average interpolation is still 0.4 dB ahead; the 3D segmentation is a threshold, not a learned model; CPU training takes about 25 minutes per run.
- Next steps: more patients, a temporal smoothness penalty during training, and a shared network across patients so that one heart can help another.

7. Conclusion

We built and tested a per-patient neural field of the beating heart. Sine activations are essential (+6.6 dB over ReLU), and so is the way the coordinates enter the network: scaling time and slice position is our own contribution and is what makes the field work between samples. The field is level with linear interpolation, better in fast contraction, and combined with it beats it, while holding the heart with 16× to 86× fewer numbers. From one network we obtain a continuous 4D heart whose ejection fraction agrees with the expert labels within five points. All code, results and data instructions are in the project repository and run with one command.

References

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